MILITARY SPECIFICATION

INDICATOR, VERTICAL AAU-9/A, INTEGRALLY LIGHTED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for one type of integrally lighted vertical velocity indicator, range 0 to 6,000 feet per minute climb and descent.

2. APPLICABLE DOCUMENTS

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-C-5541	Chemical Conversion Coatings On Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series. General Specification For
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification For The Design of
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses And Lighting Wedges

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610
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MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

FEDERALS — Colors

FED-STD-595

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-859	Standard Calibration Table For Aeronautical Pressure Measuring Equipment
MIL-STD-889	Dissimilar Metals
MS28105	Cover Glass, Aircraft Instrument Dial
MS33549	Case, Instrument, 2-3/4 Dial, With Sump, Standard For
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33649	Bosses, Fluid Connection-Internal-Straight Thread

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

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(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (See 4.4 and 6.4).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standards parts. In any case, commercial screws, bolts, nuts, and cotter pins, having suitable properties may be used with prior approval provided: They can be replaced by the standard parts (MS or AN) without alteration. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. A list of all such parts is submitted to contracting activity for approval.

3.2.2 Standard parts. With the exceptions in 3.2.1, MS and AN standards parts shall be used where they suit the purposes. They shall be identified on the drawings by their part numbers.

3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.3.2 Toxic and corrosive fumes. Materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.3.3 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of the specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic or environmental conditions shall be avoided.

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3.3.4 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.3.5.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.3.6.1 Protection of nonferrous materials. Nonferrous materials contained within hermetically sealed enclosures shall be considered suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment and anodizing of aluminum alloys parts will not be applicable for parts within hermetically sealed enclosures.

3.3.7 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.4 Design and construction. The indicator shall conform to figure 1 with the design being reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. Special tools and commercial standard tools are defined in MIL-S-8512. Special tools shall be subjected to the approval of the contracting activity.

3.4.1 Indicator constructed. Indicators shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.4.2 Performance. The indicator shall be capable of meeting the test requirements specified in Section 4 when subjected to the following conditions or any natural combination thereof:

a. Temperature. Temperature over the range of -65°C to 71°C.

b. Vibration error. Vibration with a double amplitude of between .003 and .005 inch at frequencies from 5 Hz to 50 Hz.

c. Vibration endurance. Vibration endurance at a double amplitude between 0.018 to 0.02 inch with the frequency varied from 500 to 3,000 cpm uniformly.

d. Humidity. Relative humidity up to 100 percent including conditions wherein condensation takes place in the form of both water and frost.

e. Fungus. Fungus growth as encountered in tropical climates.

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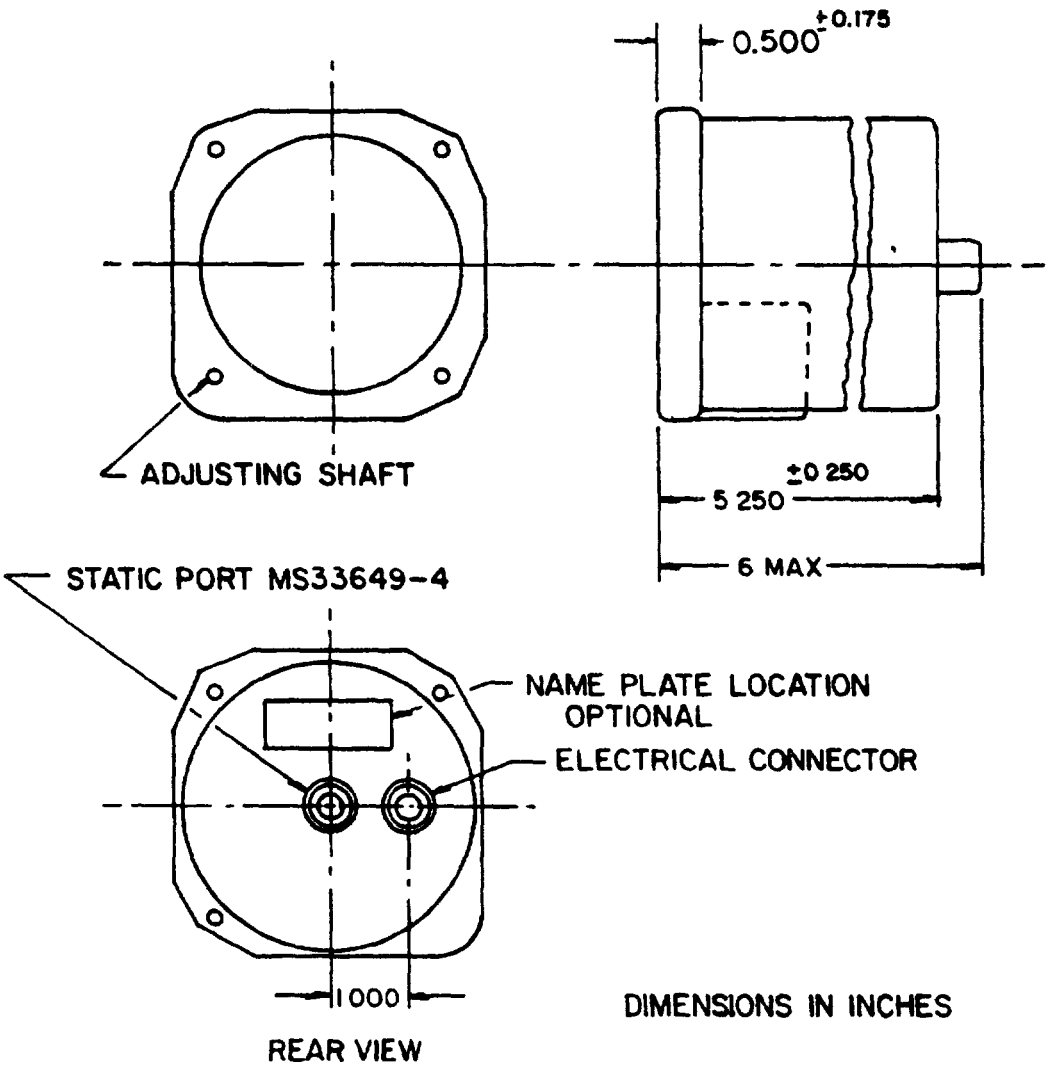


FIGURE 1 Case.

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TABLE I. Tolerance.

Ind Test Setting	Alt Interval	Tolerance	
		Rm Temp Ascent and Descent	Low & High Temp Ascent and Descent
fpm	ft	fpm	fpm
500	1,000 to 1,500	100	100
1,000	1,000 to 2,000	200	200
2,000		300	300
3,000	2,000 to 4,000	300	300
4,000		400	400
5,000		500	500
2,000	15,000 to 17,000	300	300
4,000		400	400
2,000	28,000 to 30,000	300	300
4,000		400	400

f. Salt fog. Exposure to salt sea atmosphere.

g. Altitude. Altitude in the range of sea level to 50,000 feet (3.436 inches Hg).

h. Leakage. Leakage not to exceed 0.025 inch Hg per minute when subjected to an absolute pressure of 5 inches Hg.

i. Scale error. Scale error not to exceed the limits specified in Table I.

j. Pointer lag. Pointer lag shall not be less than 7 seconds or more than 12 seconds when subjected to the test specified in 4.6.7.

k. Overpressure. Overpressure within the range specified in 4.6.11.

l. Position error. Position error which shall not exceed 50 fpm.

m. Magnetic effect. Magnetic effect not to exceed 1° in a magnetic field intensity of 0.17 to 0.19 oersted.

n. Extended range. Rate of climb of 30,000 fpm and rate of descent of 30,000 fpm shall not cause degradation of 0 to $\pm$6,000 fpm range.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

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3.6 Case. The case shall consist of a body and bezel ring and shall conform to MS33549 except as shown on figure 1. The case shall be airtight except for the static pressure connection.

3.6.1 Body. The body shall be made of nonferrous low density metal, shall be uniform in texture, and shall have a smooth surface with durable black finish. The mounting flange portion of the body shall have a dull finish.

3.6.2 Bexel ring. The bezel ring shall be made of nonferrous, low density metal and shall have a durable dull black finish. The bezel ring shall be held in place by means of screws. Provisions shall be made for the replacement of the cover glass by removal of the bezel ring.

3.6.3 Cover glass. The cover glass shall conform to MS28105 and shall be properly sealed. The cover glass shall be replaced by removal of the bezel ring which is held in place by screws properly secured with lock washers. If the cover glass is used as part of the lighting system, its size and shape may deviate from the dimensions specified in MS28105 to the extent required for proper lighting design. All reflecting glass surfaces shall be provided with a reflection reducing coating in accordance with MIL-C-14806.

3.6.4 Dial. The dial shall be as shown on figure 2.

3.6.4.1 Letters and numerals. The dial letters and numerals shall conform to MS33558.

3.6.5 Pressure fitting. Static pressure fitting shall be as shown on figure 1.

3.6.6 Plug. During shipment, the static pressure connection shall be provided with a vented plug of a material and design acceptable to the contracting activity. The vent hole shall be 0.063 $\pm$0.016 inch in diameter.

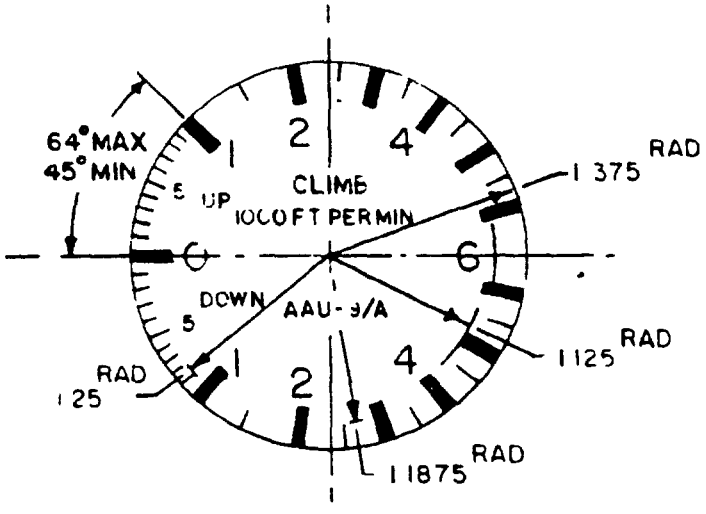
3.7 Operating mechanism. The operating mechanism shall be of the leak type. The operating mechanism shall be sufficiently rugged to prevent small distortion of the case from affecting the accuracy of the instrument.

3.8 Zero setting system. A zero setting system, controlled by a slotted setting shaft or screw shall be provided that will permit the instrument to be set to indicate zero. The use of the zero setting system shall not change the calibration of the instrument.

3.9 Visibility of dial. The pointer, numerals, at least 0.063 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 200 with a perpendicular to the dial and whose small diameter is the aperture of the instrument cas

3. 10 Pointer. The pointer shall conform to MS33585-3. It shall be light and sufficiently rigid to prevent oscillation under vibration, and shall be firmly attached to the mechanism. Suitable stops shall be incorporated within the mechanism to limit the pointer movement. The pointer shall rotate to the 6,000 foot graduation $\pm$ 0.063 inch on both ascending and descending scales.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES FRACTIONS ± 0156
ANGLES ± 0° 15'

FIGURE 2 Digl

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3.11 Electrical connector. The electrical connector shall be a Deutsch Connector DSM5600-3P, or equal, and located as shown on figure 1.

3.12 Lighting. The indicator shall be integrally lighted with two parallel lighting circuits, a white lighting circuit and a red lighting circuit as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. In the event there is any conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. The lighting shall be subject to approval by the contracting activity. Lighting shall operate on 5V ac or dc. The integral lighting shall be comparable to, and in brightness balance with, a "prime standard" indicator.

3.12.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: a minimum of 20 areas covering the entire face of the indicator shall be measured in footlamberts for brightness level; a minimum of 10 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded overall brightness and color measurements shall be approved by the contracting activity. The standard shall be checked at least after every 50 hours of operation or 120 days, whichever occurs first, to determine any change in brightness of color. Variations greater than ±20 percent in two or more areas shall require return of the standard.

3.13 Finishes and protective coatings.

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows: Dials, small holes, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541. Where primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.13.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488 may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.14 Screw threads. The threads of all machine screws 0.060 inch diameter or larger shall conform to MIL-S-7742.

3.15 Weight. The indicator shall be weighed completely assembled, without shipping plug and mounting screws, and shall not exceed 1.75 pounds.

3.16 Markings. The following markings shall be finished in lusterless white, Color No. 37875 of FED-STD-595. The dial and pointer markings dimensions shall be as follows:

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DEUTSCH CONNECTOR DM5600-3P

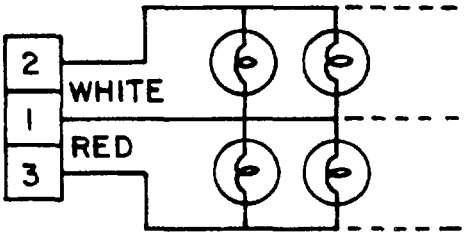


FIGURE 3. Electrical connector.

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	Height or Length Inch ± 0.016	Width of Line or Graduation Inch ± 0.005
Zero graduation		0.062
1,000 ft graduations		0.032
100 ft graduations		0.020
500 ft graduations		0.020
Arc connecting 6,000 ft		0.0312
Marks		0.0312
Numerals .5	0.125	
Numerals 0, 1, 2, 4, 6	0.313	
Lettering "CLIMB"	0.125	
Shaded part of pointer		
Lettering "UP" and "DOWN"	0.094	
Lettering "1000 ft per minute"	0.063	0.015

3.16.1 Marking. The following markings shall be finished in lusterless black, Color No. 37038, of FED-STD-595. The dimensions shall be as follows:

	Height or Length Inch ± 0.016	Width of Line or Graduation Inch ± 0.005
AAU-9/A	0.063	
Unshaded part of pointer and background of dial		

3.17 Identification of product. Equipment, assemblies, and part shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanship manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.18.1 Dimensions. Dimensions and tolerances not specified herein shall be as close as is consistent with the best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.19 Installation

3.19.1 Mounting screws. The contractor shall furnish sufficient 6-32 NC-2A round head brass machines screws, having a durable dull black oxidized or black-nickel finish. Length shall be sufficient for mounting on a panel 0.188 inch thick.

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3.19.2 Envelope. When required, the contractor shall furnish and pack with each instrument, an envelope containing mounting screws and shall be marked with the following information:

IMPORTANT

THIS ENVELOPE CONTAINS

INSTRUCTIONS, MOUNTING

SCREWS AND SPRING-NUTS

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified all inspection shall be performed in accordance with the test conditions specified in 4.3.1 of this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.3.2 Standard pressure. The standard pressure used in calibrating and testing the indicator shall be as specified in MIL-STD-859.

4.3.3 Tapping. Unless otherwise specified, the instrument shall be tested in the normal operating position and shall be lightly tapped or vibrated before a test reading is taken. Vibration shall be applied by electromechanical vibrators and shall be on the order of 0.2g's. Frequencies of less than 50 Hz shall not be used.

4.3.4 Testing. When testing several instruments in one chamber at the same time, one instrument may be selected to indicate the test rate. This test rate shall be maintained at a constant value while the test reading of all instruments are observed and noted, and while the time and altitude interval on the barometer are noted. The computed rate shall be obtained. The scale errors shall be determined.

4.4 First article inspection. The first article inspection shall consist of all the inspections specified in 4.6.

4.4.1 First articles samples. The first article inspection samples shall consist of three indicators representative of the production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.5 Quality conformance inspection.

a. Individual tests

b. Sampling plans and tests

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described in 4.6:

a. Examination of product.

b. Zero setting.

c. Case leakage.

d. Scale error at room temperature.

e. Lighting tests of MIL-L-27160 and MIL-L-25467.

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. Three indicators selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described in 4.6:

a. Individual tests (see 4.5.1).

b. Pointer lag.

c. High temperture operation.

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d. Low temperature zero error.

e. Scale error during temperature change.

f. Low temperature operation.

g. Zero error during temperature change.

h. Vibration error.

i. Sampling plan A tests of MIL-L-27160 and MIL-L-25467.

j. Overpressure.

k. Case leakage at -35°C.

4.5.2.2 Sampling plan B. Unless otherwise specified, three indicators selected at random from the first 15 items on the contract or order shall be subjected to the following tests as described in 4.6:

a. Sampling plan A (see 4.5.2.1).

b. Vibration endurance.

c. Position error.

d. Magnetic effect.

e. High altitude-low temperature.

f. High temperature exposure.

g. Humidity.

h. Fungus.

i. Salt fog.

j. Sand (fine dust).

k. Sampling plan B tests of MIL-L-27160 and MIL-L-25467.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

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4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. Prior to final assembly, the indicator shall be inspected to determine compliance with requirements of 3.18. Upon final assembly, the indicator shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, markings, weight, and dimensions. The dial shall be inspected for painting and dimensions of letters and numerals. The connector shall be inspected for bent or damaged pins. The above data shall be recorded.

4.6.2 Zero setting. The adjusting screw shall be moved throughout its range in both the UP and DOWN directions. The adjustable range of pointer movement shall be not less than 400 fpm UP and not less than 400 fpm DOWN.

4.6.3 Case leakage. The static pressure connection of the indicator shall be suitably connected to a mercury manometer and source of vacuum. An absolute pressure of 5 inches Hg (equivalent to a pressure altitude of approximately 40,000 feet) shall be applied to the static pressure connection at a rate not exceeding 2,000 fpm. With the source closed off, the difference between mercury level in the manometer during a period of 1 minute shall not exceed 0.025 inch Hg. This procedure shall be repeated with a pressure of 10 inches Hg applied. The same tolerance shall apply.

4.6.3.1 Case leakage at -35°C. The test specified in 4.6.3 shall be repeated except that the temperature shall be -35°C. The indicator shall be placed in a chamber, cooled to approximately -35°C and allowed to remain at the -35°C temperature for a period of not less than 3 hours. While at -35°C, the case leakage test shall be conducted. The difference in mercury levels shall not exceed 0.025 inches Hg in 1 minute.

4.6.4 Scale error at room temperature. The indicator, with the pointer carefully set to zero, shall be placed in an altitude chamber. A mercury barometer shall be connected to the chamber. The pressure shall be reduced at a rate such that the pointer indicates an ascent of 500 fpm. While the pointer indicates 500 fpm, the time required for the mercury barometer to change from a pressure corresponding to 1,000 feet altitude to that of an altitude of 1,500 feet shall be determined by a stop watch. The pressure shall then be increased at such a rate that the pointer will indicate a descent of 500 fpm and the time required for the barometer to change from 1,500 feet altitude to 1,000 feet altitude determined. This procedure shall be continued for the test rates and altitudes shown in table I. The scale error shall not exceed the tolerances listed in table I.

4.6.5 Lighting. The indicator shall be subjected to the individual lighting tests of MIL-L-27160 and MIL-L-25467.

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4.6.6 Pointer lag. The static pressure connection of the indicator shall be suitably connected to a water manometer and sources of vacuum and pressure. A vacuum shall be carefully appplied to the static pressure connection. The vacuum shall be sufficient to cause the pointer to indicate a rate of descent of approximately 2,500 fpm when the vacuum is instantaneously released by opening the static pressure tubing to the atmosphere. The indicator shall be tapped during the excursion of the pointer through zero. A stop watch shall be started at the instant the pointer passes the 2,000 fpm (descent) graduation and stopped when the pointer passes the 200 fpm (descent) graduation. The time shall be recorded. The procedure shall be repeated, except that a slight pressure shall be carefully applied to the static pressure connection and then released. The time shall be determined for the indication to change from 2,000 fpm climb to 200 fpm climb. The lag of the pointer shall be not less than 7 seconds nor more than 12 seconds. During this test and at any time after the vacuum (or pressure) is released, the pointer shall move smoothly without excessive friction.

4.6.7 High temperature operation. The indicator shall be subjected to a temperature of 71°C for a period of 3 hours. After completion of the 3 hour period and while still at the high temperature, the indicator shall be subjected to the tests for scale error. The indicator shall meet the requirements specified for scale error at room temperature except that the tolerance listed in table I shall be used. During and after this test the setting system shall operate satisfactorily.

4.6.8 Low temperature.

4.6.8.1 Low temperature zero error. Before starting this test, the pointer shall be carefully set to zero at room temperature. The indicator shall then be placed in a test chamber and cooled to approximately -54°C within any length of time between 15 minutes and 5 hours and allowed to remain at the temperature for a period of not less than 3 hours. With the test chamber at atmospheric pressure, a low temperature zero error during the cooling period and during the 3-hour period at -54°C, shall not exceed 150 fpm.

4.6.8.2 Scale error during temperature change. At intervals of 1 hour during the cooling period and during the 3-hour period at -54°C, the indicated rates of the indicator corresponding to the test rates of 2,000 fpm climb and descent in the standard altitude interval between 2,000 and 4,000 feet, shall be obtained. These indicated rates, corrected for the corresponding zero errors shall not exceed the values listed in table I.

4.6.8.3 Low temperature operation. After a temperature of -54°C has been maintained within the chamber for 3 hours, and while still at -54°C, the indicator shall be subjected to the test specified for scale error at room temperature. The indicated rates, corrected for the low temperature zero error, at the end of the 3-hour period at -54°C shall not exceed the values listed in table I. During and after this test the setting system shall operate satisfactorily.

4.6.8.4 Zero error during temperature change. After the low temperature scale error test has been conducted, the indicator shall be allowed to return to room temperature at atmospheric pressure by immediate removal from a temperature of -54°C to room temperature. While the temperature of the indicator is increasing, the indication shall be observed for a period of 0.500 hour, at 1 minute intervals for the first 5 minutes and at 5 minute intervals thereafter. During the first 5 minute interval the maximum reading shall not exceed 300 fpm; thereafter, the maximum reading shall not exceed 200 fpm.

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4.6.9 Vibration error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane, and with the plane of the mounting lugs of the indicator vertical. For this test, the pointer shall be set at zero. No pressure or vacuum shall be applied durinig the test. The indicator shall be subjected to vibration with in an amplitude between 0.003 and 0.005 inch at frequencies from 5 to 50 Hz in order to determine that the natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated, the maximum range of pointer oscillation at any time shall not exceed 50 fpm. The pointer variation from the zero position shall not exceed 50 fpm.

4.6.10 Sampling Plan A tests of Mil-L-27160 and Mil-L-25467. The indicator shall be subjected to and pass all Sampling Plan A tests of Mil-L-27160 and Mil-L-25487.

4.6.11 Overpressure. With the indicator in an altitude tests chamber after the pointer has been carefully set to zero, the pressure shall be decreased at a rate of 60,000 fpm until a pressure corresponding to 50,000 feet is reached. Five minutes later the indication of the pointer shall be within 100 fpm of zero. The pressure shall then be increased at a rate of 60,000 fpm until atmospheric pressure is reached. Five minutes later the indication of the pointer shall be within 100 fpm of its original indication.

4.6.12 Vibration endurance. The indicator shall be mounted on a vibration stand as specified for the vibration error test with the pointer set at zero. The indicator shall then be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 500 to 3,000 cpm and return once each hour during this test. The difference between the indication of the pointer at the beginning and at the end of the 3-hour vibration period shall not exceed 50 fpm. In addition, no looseness in the mechanism nor damage to any part of the indicator shall result from the vibration.

4.6.13 Position error. The pointer shall be set to zero. With atmospheric pressure applied, readings shall be taken while the instrument is being tapped lightly in each of several different positions. The change indication shall not exceed 50 fpm.

4.6.14. Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from and magnetically east or west from the center of a short bar magnetic compass. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass shall not exceed 1°.

4.6.15 High altitude-low temperature. The indicator shall be placed in an altitude chamber maintained at a temperature of -65°C and the pressure reduced to simulate an altitude condition of 50,000 feet (3.436 inches Hg absolute pressure) for a period of 4 hours. The indicator shall then be maintained at a temperature of -65°C atmospheric pressure, for an additional period of 44 hours. After returning to room temperature, the indicator shall be carefully examined for any evidence of damage. No damage to any part of the indicator shall result from this test.

4.6.16 High temperature exposure. The indicator shall be subjected to a temperature of $71^{\circ} \pm 2^{\circ}$C for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the indicator shall operate without indication of sticking or erratic action. After the chamber has been returned to room temperature and the indicator has saturated at room temperature for a minimum period of 4 hours, it shall meet the individual test requirements. There shall be no damage which would adversely affect subsequent operation.

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4.6.17 Humidity. The indicator shall be subjected to the humidity test in accordance with MIl-STD-810, Method 507.1 procedure I. The indicator shall then be subjected to and shall meet the individual tests.

4.6.18 Fungus. The indicator, including applicable external connectors, shall be subjected to the fungus resistance test in accordance with MIL-STD-810, Method 508.1 procedure I. At the end of the test period the indicator shall be examined to ascertain that no fungus growth has occured. The indicator shall then be subjected to and shall meet the individual tests.

4.6.19 Salt fog. The indicator shall be subjected to a salt fog test in accordnce with MIL-STD-810, Method 509.1 procedure I, for a period of 48 hours. At the end of the 48-hour period, there shall be no damage or deterioration to any part of the indicator. The indicator shall then be subjected to and shall meet the individual tests.

4.6.20 Sand (fine dust). The indicator shall be subjected to a sand (fine dust) test in accordance with MIL-STD-810, Method 510.1, procedure I. At the end of the test the indicator shall then be subjected to and shall meet the individual tests.

4.6.21 Sampling Plan B tests of Mil-L-27160 and Mil-L-25467. The indicator shall be subjected to and pass all Sampling Plan B tests of Mil-L-27160 and Mil-L-25467.

4.6.22 Lighting. The indicator shall be subjected to and shall meet the sampling plan A lighting tests of MIL-L-27160 and MIL-L-25467.

4.6.23 Lighting. The indicator shall be subjected to and shall meet the sampling plan B lighting tests of MIL-L-27160 and MIL-L-25467.

4.7 Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or industrial as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

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5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords protection from domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vertical velocity indicator is intended for use on aircraft to indicate the vertical component of the aircraft speed based on the rates of change of atmospheric pressure. The vertical velocity range is from 0 to 6,000 fpm and 0 to -6,000 fpm.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Indicators which have been subjected to the sampling tests may be delivered of contract provided:

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(1) The sample has been refurbished.

(2) The item meets the individual tests.

(3) The sample is representative of production indicator currently being accepted.

c. Whether sampling plan B is to be omitted.

d. Level of packaging and packing required.

e. Special shipment marking (see 5.3)

6.3 Definitions

6.3.1 Tapping. The phrase tapped lightly is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Amplitude. Whenever the word amplitude is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3.3 Test reading. The indication of the instrument during the tests.

6.3.4 Test rate. The test point settings shown in column 1 of table I at which the instrument is to be tested.

6.3.5 Computed rate. The rate computed by dividing the altitude interval (2,000 feet as measured by a manometer) by the time in seconds (required for the change in altitude) and multiplying by 60.

$$\text{Computer rate} = \frac{\text{Altitude interval X 60}}{\text{Time in seconds}}$$

6.3.6 Computer rate correction. The algebraic difference between the test rate and the computed rate.

Example 1	Test Rate	+2,000
	Computed Rate	+1,925
	Computed Rate Correction	+75
Example 2	Test Rate	+2,000
	Computed Rate	+2,025
	Computed Rate Correction	-25

6.3.7 Indicated rate. The reading of the instrument as computed herein to correspond to a test rate. The indicated rate of the instrument shall be computed by adding algebraically the test reading of the instrument and the computed rate correction.

6.3.8 Scale error. The difference between the test rate and the indicated rate.

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6.3.9 Refurbished. Refurbished means that the indicator has been completely overhauled, that all component parts meet aircraft parts standards, and that the indicator meets all requirements of a new indicator.

6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.5 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement, STANAG 3320 and ASCC Air Standard 10/8. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement, concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

6.6 Subject term (key word) listing.

AAU-9/A
Indicator
Integrally
Vertical Velocity
0-600 feet/min

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Review activity
AIR FORCE -11

Preparing activity:
AIR FORCE -71

Project number:
6610-F182

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1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
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	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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